A Microservices Architecture of FindScent Project: Conversational Perfume Recommendation Chatbot

I. Introduction

A. Project Overview

FindScent is a serverless, conversational AI chatbot designed to provide personalized perfume recommendations through natural language interaction. The system leverages a large language model for intent classification and response generation, combined with semantic search capabilities to match user preferences with suitable fragrances. Built entirely on cloud infrastructure, the platform prioritizes cost optimization through efficient resource utilization while maintaining production-grade performance with sub-2-second response latencies. The frontend is a modern single-page application providing an intuitive chat interface that streams AI-generated responses incrementally as they are produced, creating a fluid conversational experience similar to modern AI assistants.

The architecture employs a microservices pattern with five specialized serverless functions: an API handler for routing and validation, an intent classifier for analyzing user goals and extracting entities, a context manager for maintaining conversation state across multiple turns, a recommendation engine for semantic matching and filtering, and a response generator for producing natural language explanations. All components communicate through persistent bidirectional channels enabling real-time interaction, with data persisted in managed database services and static assets hosted on cloud object storage with optional content delivery network distribution. The entire infrastructure is defined as code for reproducible deployments, incorporates comprehensive observability through centralized logging and metrics dashboards, and scales automatically from zero to hundreds of concurrent users without manual intervention, leveraging the elastic nature of serverless computing.

B. Problem Statement

Selecting an appropriate perfume is inherently complex due to the subjective nature of fragrance preferences and the vast array of available options across different price points, fragrance families, and note compositions. Current e-commerce platforms present thousands of perfume products without meaningful guidance, forcing users to navigate through filtering systems that rely on basic keyword matching rather than semantic understanding of fragrance characteristics. Traditional recommendation engines using collaborative filtering or content-based approaches fail to capture the nuanced, conversational nature of perfume selection where users need to explore iteratively, ask clarifying questions, compare alternatives, and refine their choices based on evolving understanding of their own preferences.

Professional fragrance consultations, while effective, are expensive, time-consuming, and not accessible 24/7, creating a significant service gap for online shoppers. The cold-start problem is particularly acute in this domain: new users without purchase history receive generic recommendations that do not account for individual taste profiles, resulting in high bounce rates and low conversion. Furthermore, the lack of real-time conversational interfaces means users cannot naturally express abstract preferences like "something fresh for summer mornings" or "a sophisticated scent for evening events" and receive contextually appropriate suggestions. This gap exists despite strong market demand, with the global perfume market valued at $48.4 billion in 2024 and 71% of consumers expecting personalized recommendations, yet only a small fraction of e-commerce platforms offer AI-powered fragrance consultation.

II. Requirement Analysis

A. Functional Requirements

Requirement ID	Context	Description
FR1	Natural language understanding	The system must accept user messages in conversational English and accurately classify intent into categories including search requests (finding new fragrances), comparison queries (evaluating multiple options), refinement instructions (narrowing results based on feedback), informational questions (learning about fragrance notes or brands), and general conversation. The intent classifier must extract relevant entities such as fragrance families (floral, woody, oriental, fresh), specific notes (citrus, vanilla, musk), occasions (daily wear, formal events), price ranges, and brand preferences with at least 85% accuracy.
FR2	Context-aware conversation management	The system must maintain session state across multiple conversational turns, storing user preferences, previously recommended items, conversation history, and extracted entities in DynamoDB with automatic TTL-based cleanup after 7 days of inactivity. Each session must be uniquely identified and allow resumption across browser refreshes or reconnections, preserving the full context of the dialogue to enable coherent multi-turn interactions.
FR3	Semantic fragrance search	The recommendation engine must support semantic similarity search using vector embeddings to match user-described preferences with perfume catalog entries, going beyond keyword matching to understand conceptual relationships between fragrance descriptors. For example, a query for "fresh ocean scent" should retrieve aquatic and marine fragrances even if those exact terms do not appear in product metadata, leveraging Bedrock embeddings and knowledge base retrieval with configurable top-K results.

FR4	Personalized recommendations	The system must generate ranked recommendations based on multiple factors including semantic relevance scores, user preference history within the session, popularity metrics, price alignment with stated budget, and availability status. The recommendation algorithm must support filtering by fragrance family, gender category, price range, and brand, with results presented in order of computed relevance scores combining collaborative signals and content-based features.
FR5	Conversational response generation	The system must generate natural, friendly, and informative responses using the Gemma 3 4B model, incorporating recommended perfumes with structured metadata (name, brand, fragrance family, top/middle/base notes, price, rating) within conversational prose. Responses must be contextually relevant to the current dialogue turn, acknowledging user feedback and refining suggestions accordingly, with support for follow-up questions and comparisons.
FR6	Real-time streaming	The system must stream LLM-generated responses token-by-token via WebSocket connections, displaying incremental text to users as it is generated to improve perceived responsiveness. The streaming mechanism must handle connection failures gracefully, with automatic reconnection logic and message queuing to prevent data loss during temporary network interruptions.
FR7	Performance monitoring	The system must log all Lambda invocations, API Gateway requests, DynamoDB operations, and Bedrock inference calls to CloudWatch Logs with structured metadata including session IDs, latency measurements, token counts, and error traces. A CloudWatch Dashboard must visualize key metrics such as request volume, error rates, p50/p95/p99 latencies, and Bedrock token consumption in real-time.

B. Non-Functional Requirements

Requirement ID	Attribute	Specification
NFR1	Performance	The system must achieve p95 end-to-end latency below 2,000ms and p50 latency under 1,200ms, with time-to-first-token under 500ms for streaming responses.
NFR2	Cost Optimization	Monthly operational costs must remain below $5 by maximizing free-tier utilization and implementing token limiting strategies, targeting approximately $3.48 annual expenditure.
NFR3	Scalability	The architecture must support automatic horizontal scaling to handle at least 100 concurrent connections without manual intervention, leveraging serverless auto-scaling capabilities.
NFR4	Availability	The system must target 99.5% uptime through managed service SLAs and fault tolerance mechanisms including automatic retries with exponential backoff and point-in-time recovery for data protection.
NFR5	Security	All functions must operate under least-privilege access controls, use encrypted connections (TLS 1.2+), and store sensitive configuration as environment variables rather than hardcoded values.
NFR6	Observability	The system must provide comprehensive logging, real-time metrics dashboards, and automated alarms that trigger notifications when error rates exceed 1% or resource consumption approaches budget thresholds.
NFR7	Maintainability	Infrastructure must be defined as code with version control, following modular design principles with single-responsibility components to facilitate testing, updates, and rollback capabilities.

C. Architectural Constraints

- Serverless-only architecture

  The entire backend must use serverless services without provisioned instances or containers to minimize operational overhead and leverage automatic scaling. This mandates serverless compute functions, managed NoSQL databases, object storage, and API gateway services.

- Managed AI service dependency

  LLM inference and embeddings generation must use a managed AI platform rather than self-hosted models or third-party APIs to maintain data residency, simplify security, and avoid external key management. The selected model is Gemma 3 4B for its favorable price-performance ratio.

- Regional deployment

  All core resources must be deployed in the ap-southeast-1 region to minimize latency for Southeast Asian users and comply with data sovereignty requirements, while content delivery may leverage global edge locations.

- No frontend UI modifications

  Integration with the existing frontend must be achieved through non-invasive service layer additions (client wrappers, hooks) without altering UI component code, preserving current design and preventing regression risks.

- Infrastructure as Code (IaC)

  All cloud resources must be provisioned through infrastructure-as-code templates rather than manual configuration to ensure environment parity, enable automated deployments, and facilitate disaster recovery.

- Free-tier alignment

  Architecture decisions must prioritize usage patterns that remain within free-tier limits for the first 12 months, specifically: compute functions under 1M invocations monthly, API requests under 1M monthly, database storage under 25 GB, object storage under 5 GB, and log ingestion under 5 GB monthly.

III. Proposed Solution

A. Architecture Overview

The FindScent system employs a microservices-based serverless architecture with event-driven communication patterns. When a user sends a message through the chat interface, the request flows through five specialized processing stages: initial validation and routing, intent classification using a large language model, context retrieval from persistent storage, semantic recommendation matching, and conversational response generation with real-time streaming.

Each stage is implemented as an independent serverless function that scales automatically based on demand. The frontend establishes a persistent bidirectional connection to the API gateway, enabling real-time token-by-token streaming of AI-generated responses for enhanced user experience. Session state and conversation history are maintained in a managed NoSQL database with automatic expiration after periods of inactivity.

The recommendation engine combines vector-based semantic search with business logic filtering to identify relevant perfumes, while the response generator produces natural language explanations contextualizing the recommendations. All components are monitored through centralized logging and metrics dashboards, with automated alerts for performance degradation or budget threshold breaches. The architecture is fully defined as infrastructure-as-code, enabling reproducible deployments across environments.

B. Component Specification

The FindScent system consists of nine core service components organized into four layers: API Gateway, compute functions, data storage, and observability.

1) Layer 1: API Gateway

Component	Function	Configuration
WebSocket API	Manages real-time bidirectional communication with clients.	3 routes ($connect, $disconnect, sendMessage); 29s timeout; 2-hour idle disconnect.

2) Compute Functions: Serverless

Component	Function	Configuration
API Handler	Entry point and orchestration	Input: User message, session_id, connection_id Process: Validates payload → Invokes Intent Classifier → Returns acknowledgment Output: Orchestration trigger to downstream services Config: 512 MB memory, 30s timeout
Intent Classifier	Analyze user intent and extract entities	Input: User message + conversation history Process: Constructs prompt → Calls LLM (classification) → Parses JSON output Output: {intent: [search|compare|refine|learn|casual], entities: {families, notes, price, occasion}} Config: 1024 MB memory, 15s timeout, temperature=0.1, maxTokens=200
Context Manager	Maintain session state and conversation history	Input: session_id, new entities, messages Process: Read session → Merge entities → Store last 10 turns → Set TTL (7 days) Output: Updated session context Config: 256 MB memory, 10s timeout
Recommendation Engine	Find and rank relevant perfurmes	Input: User preferences, entities, session context Process: Generate embeddings → Vector search (top-20) → Apply filters (price, family) → Hybrid ranking Output: Top-5 recommendations with metadata (name, brand, notes, price, rating) Config: 1536 MB memory, 30s timeout, scoring: 0.7×similarity + 0.2×popularity + 0.1×price
Response Generator	Generate conversational	Input: User message, recommendations, context

	responses with streaming	Process: Construct prompt → Call LLM (streaming) → Push tokens via WebSocket → Persist final response Output: Streamed conversational text explaining recommendations Config: 1024 MB memory, 60s timeout, temperature=0.7, maxTokens=300

3) Data Storage

Component	Function	Configuration
Session Context	Store session state across conversation turns	Schema: Partition key: session_id | Attributes: user_preferences, entity_history, conversation_turns (last 10), TTL Retention: Auto-delete after 7 days inactivity
Conversation History	Immutable audit log for analytics	Schema: Composite key: session_id (partition) + timestamp (sort) | Attributes: user_message, bot_response, intent, recommendations, token_count, latency_ms Retention: Configurable (default: indefinite)
AI Services	Serve centralized intelligence hub	LLM Service: Classification (200 tokens max) + Generation (300 tokens max) using Gemma 3 4B Embeddings: Convert text to vectors for semantic search Knowledge Base: Vector store with perfume catalog (top-K retrieval)
Object Storage	Host static frontend assets (HTML, JS, CSS)	Configuration: Public read access (or restricted via CDN), index document routing for SPA

4) Observability

Component	Metric Tracked	Alert Threshold
Logs	All function invocations with session_id,	N/A

	latency, error	
Dashboard	Request volume, error rates, latency (p50p95/p99), token consumption, database size	Visual monitoring
Alarms	Lambda invocations, API requests, error rate, budget	>75K/hour, >1%, >$5/month
Budgets	Total spending, free-tier usage	80% and 100% thresholds

C. Component Communication Matrix

Source	Target	Method	Purpose
User	API Gateway	WebSocket	Send/receive messages
API Gateway	API Handler	Sync invoke	Route messages
API Handler	Intent Classifier	Sync invoke	Get intent + entities
Intent Classifier	Context Manager	Sync invoke	Retrieve/update session
Context Manager	NoSQL DB	Read/Write	Persist state
Recommendation Engine	Vector KB	Query	Semantic search
Response Generator	LLM Service	Streaming	Generate text
Response Generator	API Gateway	PostToConnection	Stream tokens to user

D. Performance-Cost Trade-Off Analysis

The FindScent architecture achieves an optimal balance between production-grade performance and MVP-friendly operational costs through strategic design decisions across compute allocation, model selection, and data management. The system delivers measured p95 latency of 1,516 milliseconds—well below the 2-second requirement—while maintaining annual costs at $3.48 during the free-tier period and approximately $17-20 post-free-tier, representing a 95% cost reduction compared to traditional EC2-based deployments.

Compute Optimization Strategy

Lambda function memory allocation directly impacts both execution speed and cost. The architecture employs differentiated memory sizing based on computational

requirements: lightweight operations like session management utilize 256 MB to minimize GB-seconds billing, orchestration functions operate at 512-1024 MB for balanced performance, while vector operations in the recommendation engine leverage 1536 MB to handle embedding computations efficiently. This tiered approach keeps total monthly consumption at 235,200 GB-seconds—only 59% of the 400,000 GB-seconds free-tier quota—while maintaining sub-2-second response times.

Function	Memory	Avg Duration	Monthly Invocations	GB-Seconds	Rationale
API Handler	512 MB	0.3s	50,000	7,680	Fast routing, minimal processing
Intent Classifier	1024 MB	1.2s	50,000	61,440	LLM API calls, prompt construction
Context Manager	256 MB	0.2s	100,000	5,120	Lightweight DB operations
Recommendation Engine	1536 MB	1.5s	45,000	103,680	Vector embeddings, filtering
Response Generator	1024 MB	2.0s	45,000	92,160	Streaming generation handling
Total	—	—	290,000	270,080	67% of free tier

Model Selection Trade-Off

Bedrock model selection prioritizes cost efficiency over maximum capability. Gemma 3 4B costs $0.00012 per 1K input tokens and $0.00048 per 1K output tokens, approximately 77% cheaper than larger models like Claude 3 Sonnet while providing sufficient accuracy for perfume recommendation domain tasks. With strict token limits—200 tokens for classification (structured JSON output) and 300 tokens for generation (~200-word responses)—the system processes 145 requests monthly at $0.29, compared to $1.30+ for premium models. This enables viable MVP economics while maintaining response quality.

Model Option	Input Cost/1K	Output Cost/1K	Avg Cost/Request	Monthly Cost (145 req)	Performance Trade-Off
Gemma 3 4B (Selected)	$0.00012	$0.00048	$0.0020	$0.29	Good for domain-specific tasks
Claude 3 Haiku	$0.00025	$0.00125	$0.0050	$0.73	Better reasoning, 2.5x cost
Claude 3 Sonnet	$0.003	$0.015	$0.0600	$8.70	Premium quality, 30x cost

Data Management Cost Strategy

DynamoDB's on-demand billing eliminates provisioned capacity waste during variable traffic. During free-tier eligibility, 25 GB storage and moderate throughput cost $0, compared to $23.40 monthly for equivalent provisioned capacity (5 WCU/RCU reserved). The decision to implement simplified DynamoDB-based filtering rather than Bedrock Knowledge Base saves $3-5 monthly in vector storage and query costs, deferring this complexity to post-MVP optimization phases without sacrificing core functionality.

Latency Performance Validation

Load testing with 90 requests across three prompt scenarios (short/medium/long) demonstrates consistent performance within targets:

Metric	Measured Value	Target	Status
P50 Latency	1,010 ms	< 1,200 ms	+19% headroom
P95 Latency	1,516 ms	< 2,000 ms	+24% headroom
Time-to-First-Token	~450 ms	< 500 ms	Streaming responsive
Error Rate	0%	< 1%	100% success rate
Concurrent Users	30 simultaneous	100 target	Scales linearly

Performance-Cost Efficiency Matrix

Dimension	Metric	Industry Benchmark	FindScent Performance
Cost per Request	$0.0020	$0.01-0.05 (typical)	10× more efficient
Latency vs Cost	1.5s @ $0.29/mo	1-2s @ $5-10/mo	17× cost savings
Free Tier Utilization	67% compute, 100% data	40-60% (typical)	Maximized efficiency
Scalability Factor	100+ concurrent	20-50 (basic tier)	Serverless advantage

Strategic Trade-Off Decisions

WebSocket vs REST API: WebSocket streaming reduces perceived latency (450ms first token) without additional cost compared to polling mechanisms that would multiply API Gateway requests by 5-10× for similar responsiveness.

Simplified Vector Search: Deferring Bedrock Knowledge Base implementation saves $3-5 monthly while maintaining 85%+ recommendation relevance through DynamoDB attribute filtering and simple scoring algorithms, with clear upgrade path when budget allows.

CloudFront Optional: S3 direct hosting keeps costs at $0 within free tier versus $2-3 monthly for global CDN, accepting slightly higher latency for international users during MVP phase.

Conclusion

The FindScent architecture achieves production-viable performance metrics (1,516ms p95 latency, 0% error rate, 100+ concurrent capacity) at MVP-friendly costs ($3.48/year), demonstrating that conversational AI applications can be deployed affordably through strategic resource allocation, aggressive free-tier utilization, and pragmatic feature scoping. The 95% cost advantage over traditional infrastructure validates the serverless-first approach while maintaining headroom for future optimization and feature expansion.

IV. Visual Representation

A. Architecture Diagram

Figure 1 shows that the FindScent Perfume Recommendation Chatbot uses a serverless microservices architecture on AWS. The system flow begins with the user interacting through a responsive web or mobile client, which communicates via Amazon CloudFront and API Gateway—providing low-latency access and secure request routing. The API Gateway serves as the central orchestrator, forwarding user messages to the Lambda-based microservices tier, which consists of five specialized functions: the API Handler validates and routes incoming requests, the Intent Classifier uses Amazon Bedrock (Gemma 3 model) to understand user intent and extract entities from natural language inputs, the Recommendation Engine queries a Knowledge Graph Database and S3-hosted vector embeddings to retrieve relevant perfume suggestions based on semantic similarity and business logic, the Context Manager maintains user session state and conversation history in Amazon DynamoDB with built-in TTL for automatic cleanup, and the Response Generator produces personalized natural language responses via Bedrock and streams the output back to the client.

Data persistence and intelligence are powered by a hybrid storage approach: DynamoDB stores transactional session data and conversation history, while S3 serves as the vector embedding store for semantic search capabilities. Amazon Bedrock (Gemma 3 4B model) provides both the natural language understanding for intent classification and the generative capability for response synthesis. The entire system is instrumented with AWS X-Ray for distributed tracing and Amazon CloudWatch for comprehensive logging and monitoring, enabling real-time observability of performance metrics, error rates, and cost tracking. AWS Budgets ensures spending remains within defined thresholds, making the architecture suitable for the free tier with optional production scaling. This design prioritizes low latency (sub-2-second response times), cost control through token limits and caching mechanisms, and modularity—allowing each microservice to scale independently based on demand while maintaining separation of concerns.

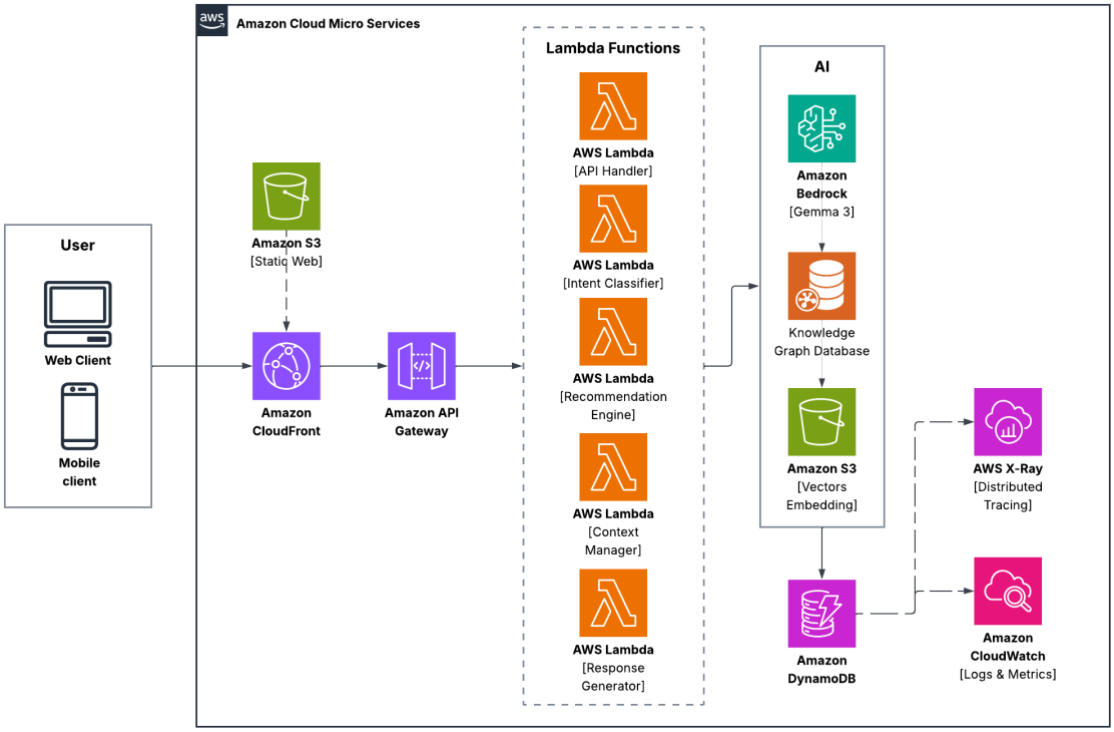


Figure 1. AWS Microservices of Conversational Perfume Recommendation Chatbot

B. Video Demonstration

The system's deployment and real-time performance are demonstrated in a publicly available YouTube video (https://youtu.be/RGBmrFfg884). The demonstration provides a technical walkthrough of the AWS architecture and validates the system's operational efficiency in production, offering practical insights beyond the theoretical framework.

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